Week 8

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval. (CLO 6) **2.** Construct a percentile bootstrap confidence interval with StatKey. (CLO 6) **3.** Predict how changing the confidence level affects the width of an interval. (CLO 6) **4.** Predict how changing the sample size affects the width of an interval. (CLO 6) **5.** Evaluate whether a bootstrap distribution is symmetric and bell-shaped enough to trust a bootstrap interval. (CLO 6) **6.** Explain why a bootstrap interval cannot correct for a non-representative sample. (CLO 1, CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Confidence Intervals Using Percentiles – Mon: StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods; Your Turn: Percentile CIs (124 of 200 students work)	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
7. Identify the parameter of interest in a research question. (CLO 8) **8.** Write null and alternative hypotheses in symbols and in words. (CLO 8) **9.** Classify a test as right-tailed, left-tailed, or two-tailed. (CLO 8) **10.** Explain the logic of a hypothesis test: assume H_0 is true and ask whether the sample data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Writing Hypotheses – rock paper scissors practice (9 of 11 wins): define the parameter, state H_0 and H_a, choose the tail; Your Turn: Write the Hypotheses (airline on-time rate, SAT prep, and more) Think about it: is 15 of 16 correct (dolphin study) convincing?	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM) Take-Home Quiz 5 (due Mon Oct 26, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus